# ML and DS Internship Practical

## Month 4 – Real-World Projects

### Practical 2: Web Scraper


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**Submitted To: -**

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Technology Used: Python

Development Environment: Visual Studio Code

# WEB SCRAPING

## 1. Introduction

The Web Scraping and Data Cleaning project is developed using Python to extract book information from the Books to Scrape website. The project uses Requests to retrieve webpage content, BeautifulSoup to parse HTML, and Pandas to clean and validate the collected data.

The final processed dataset is exported to a CSV file, which can be opened in Microsoft Excel for further analysis.

## 2. Objectives

- To automate the collection of book information from a website.
- To extract product names, prices, ratings, and availability.
- To navigate through multiple pages using pagination.
- To clean and validate the extracted data.
- To identify and remove duplicate and incomplete records.
- To export the cleaned dataset into CSV format.

## 3. Technologies Used

| Technology | Purpose |
|---|---|
| Python | Main programming language |
| Requests | Fetches webpage content |
| BeautifulSoup | Parses HTML and extracts information |
| Pandas | Cleans and validates data |
| CSV | Stores the processed dataset |
| OS Module | Creates the output directory |
| urllib.parse | Handles URLs during pagination |

## 4. Project Description

The project extracts information from the Books to Scrape practice website.

The following details are collected:

- **Product Name:** Name of each book.
- **Price:** Price of the book in pounds.
- **Rating:** Rating represented as a number from 1 to 5.
- **Availability:** Stock status displayed on the website.

The scraper automatically navigates through available pages and collects product information until the last page is reached.

Website: https://books.toscrape.com/

# 5. System Workflow

1. Send an HTTP request to the website.
2. Receive the webpage's HTML content.
3. Parse the HTML using BeautifulSoup.
4. Extract book names, prices, ratings, and availability.
5. Navigate through the website using pagination.
6. Create a Pandas DataFrame.
7. Validate and clean the extracted data.
8. Remove duplicate and incomplete records.
9. Export the cleaned dataset to a CSV file.

# 6. Data Cleaning and Validation

The project performs the following operations:

Currency Removal: Removes the pound symbol from prices.

Example: £51.77 becomes 51.77.

Price Conversion: Converts price values into numeric floating-point values.

Rating Conversion: Converts rating words into numbers.

| Rating | Numeric Value |
|---|---|
| One | 1 |
| Two | 2 |
| Three | 3 |
| Four | 4 |
| Five | 5 |

Duplicate Removal: Identifies and removes duplicate rows using Pandas.

Missing Value Handling: Checks for missing values and removes incomplete records.

Invalid Data Checks: Identifies invalid or negative prices, invalid ratings, and empty product names.

Index Reset: Resets the DataFrame index after cleaning.

# 7. Output

After successful execution, the cleaned dataset is saved at:

output/products_cleaned.csv

The output directory is created automatically if it does not already exist.

The CSV file contains the following columns:

| Column | Description |
|---|---|
| Product Name | Name of the book |
| Price | Numeric price in pounds |
| Rating | Numeric rating from 1 to 5 |
| Availability | Stock availability |

The generated file can be opened in Microsoft Excel for reviewing, sorting, filtering, and further analysis.

## 8. Testing

The project can be tested using the following scenarios:

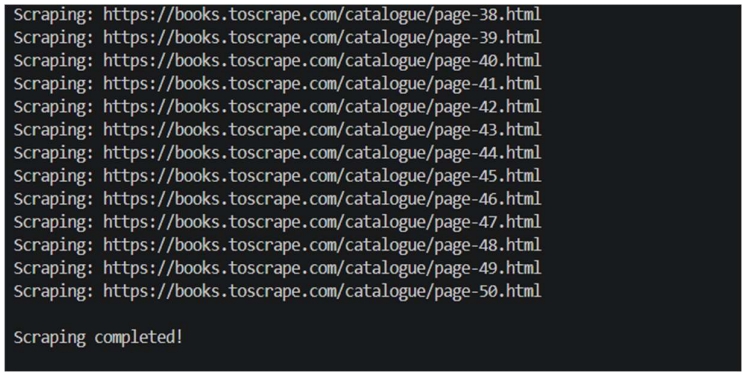
| Test Case | Expected Result |
|---|---|
| Website connection | Website response is received |
| Product extraction | Book details are collected |
| Pagination | Multiple pages are processed |
| Price conversion | Prices become numeric values |
| Rating conversion | Ratings become numeric values |
| Duplicate checking | Duplicate rows are removed |
| Missing value checking | Incomplete records are identified and filtered |
| CSV export | Cleaned dataset is saved successfully |
| Website request failure | An error message is displayed |

Actual test results should be recorded after running the program.

## 9. Limitations

- The project depends on the availability of the target website.
- Changes to the website's HTML structure may require code modifications.
- Network errors may interrupt data collection.
- Data validation is based on the rules implemented in the program.
- The scraper is designed for the Books to Scrape practice website.

## 10 .Screenshot



```
--- DATA VALIDATION ---
Records before cleaning: 1000

Missing values:
Product Name    0
Price           0
Rating          0
Availability    0
dtype: int64

Invalid prices: 0
Invalid ratings: 0
Duplicate rows: 0

Records after cleaning: 1000
Data cleaning completed!

--- FINAL REPORT ---
Total products: 1000

Column data types:
Product Name         str
Price            float64
Rating             int64
Availability         str
dtype: object
```

```
Missing values after cleaning:
Product Name    0
Price           0
Rating          0
Availability    0
dtype: int64

First 10 products:
                                                                                Product Name  Price  Rating Availability
                                                                        A Light in the Attic  51.77       3     In stock
                                                                          Tipping the Velvet  53.74       1     In stock
                                                                                  Soumission  50.10       1     In stock
                                                                               Sharp Objects  47.82       4     In stock
                                                                  Sapiens: A Brief History of Humankind  54.23       5     In stock
                                                                             The Requiem Red  22.65       1     In stock
                                                          The Dirty Little Secrets of Getting Your Dream Job  33.34       4     In stock
                The Coming Woman: A Novel Based on the Life of the Infamous Feminist, Victoria Woodhull  17.93       3     In stock
The Boys in the Boat: Nine Americans and Their Epic Quest for Gold at the 1936 Berlin Olympics  22.60       4     In stock
                                                                             The Black Maria  52.15       1     In stock

CSV saved at: output/products_cleaned.csv
```

## 11. Conclusion

The Web Scraping and Data Cleaning project demonstrates how Python can automate data collection from a website and transform raw information into a structured dataset.

By using Requests, BeautifulSoup, and Pandas, the project combines web scraping, pagination, data validation, data cleaning, and CSV export in a single workflow. It provides practical experience in Python programming and introductory data preprocessing.

## 12. Reference

Books to Scrape: https://books.toscrape.com/

This website is used for educational web scraping practice.

## 13. GitHub Repository Link: -

**https://github.com/saniya2698/Web-Scraper-Python**